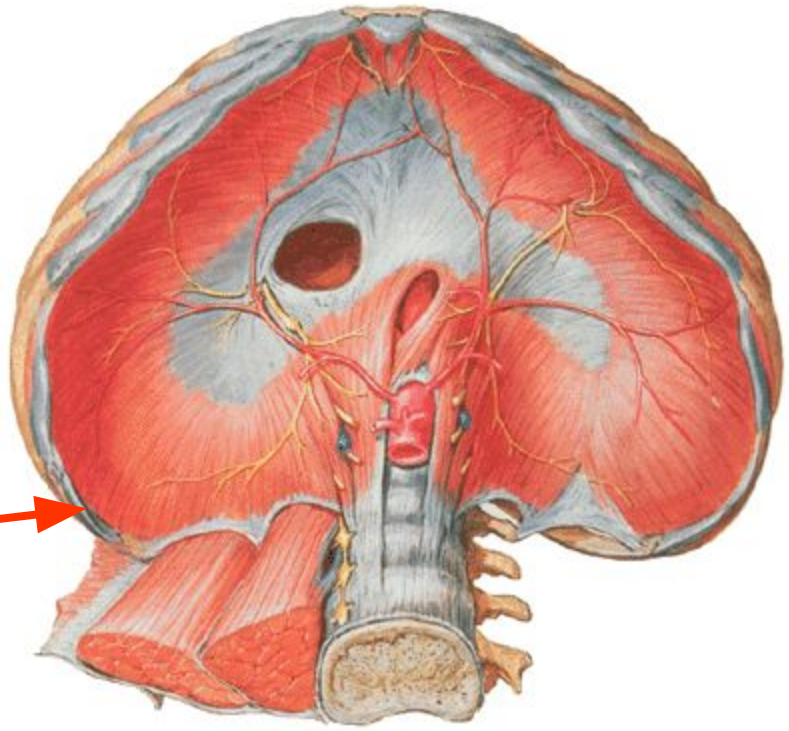
OUTLET OF THE THORAX

- **It is the broad end of thorax**
- **Surrounds the upper part of abdominal cavity**
- **Separates the thoracic from abdominal cavity by diaphragm**



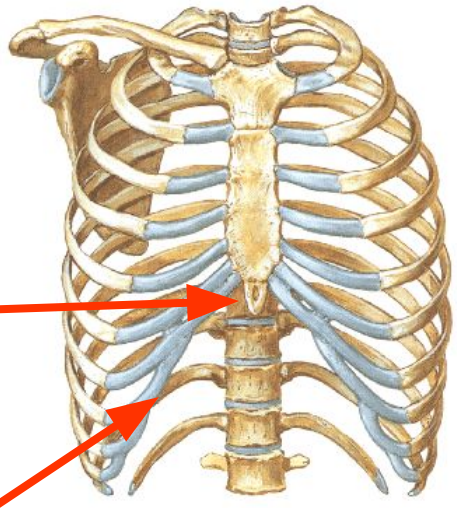
BOUNDARIES

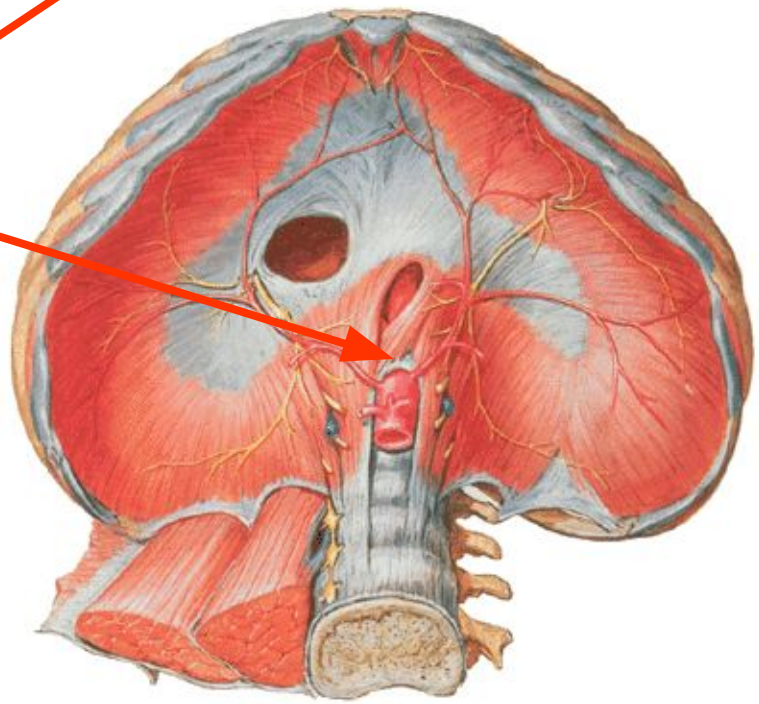
- **Anteriorly**

  Infrasternal angle between the two costal margins.
- **Posteriorly**

  Inferior surface of the body of 12th thoracic vertebra.
- **On each side**

  Cartilages of 7th-10th ribs and
  11th and 12th rib.

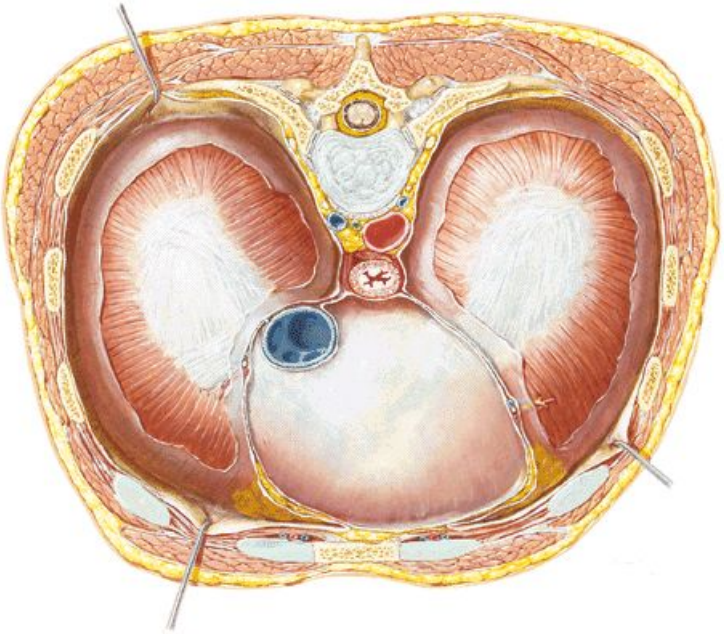


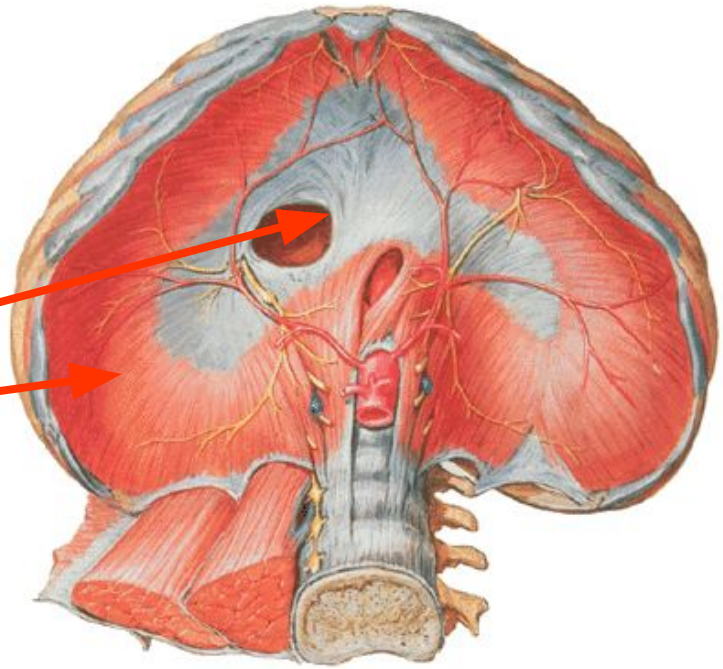


DIAPHRAGM

- Dome shaped
- Fibro-muscular sheet
- Separates thoracic and abdominal cavities
- Has right & left domes
- Chief muscle of respiration

- Composed of
  - Central Tendinous part
  - Peripheral Muscular part

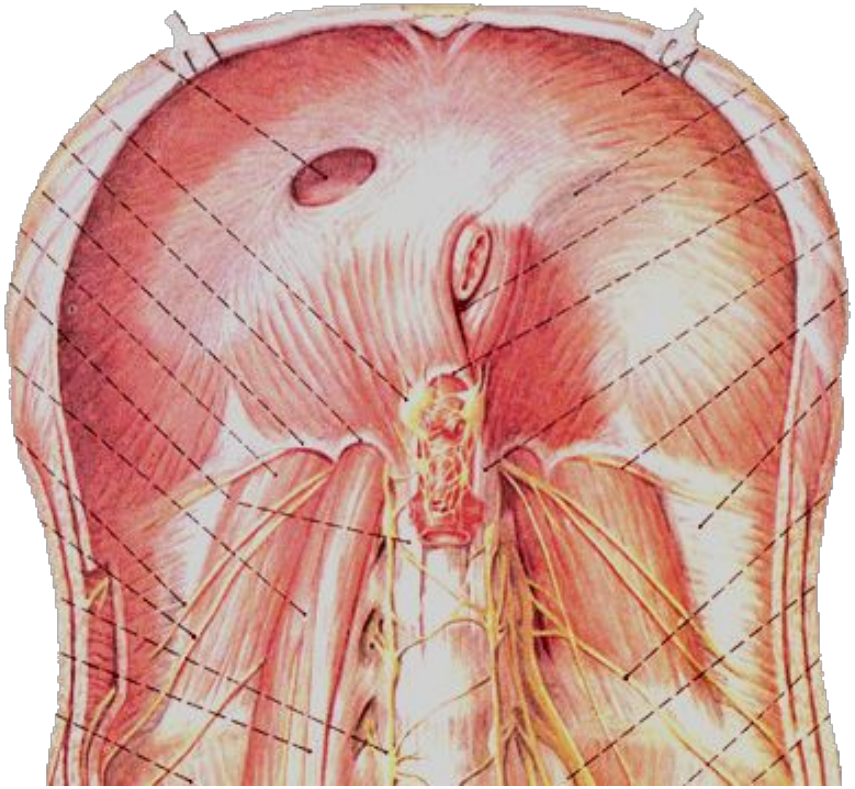




Diaphragm

Origin

- **Lumbar part:** arises by two crura from upper 2-3 lumbar vertebrae
- **Costal part:** lower six ribs and their costal cartilages
- **Sternal part**: xiphoid process



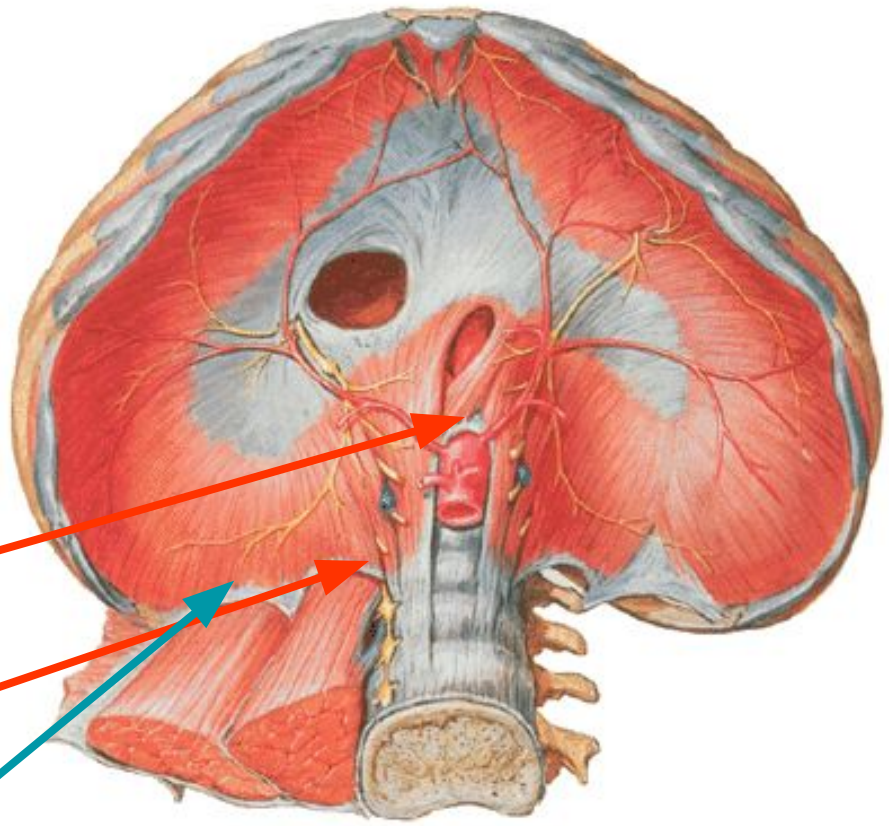
Insertion: central tendon

Vertebral crura

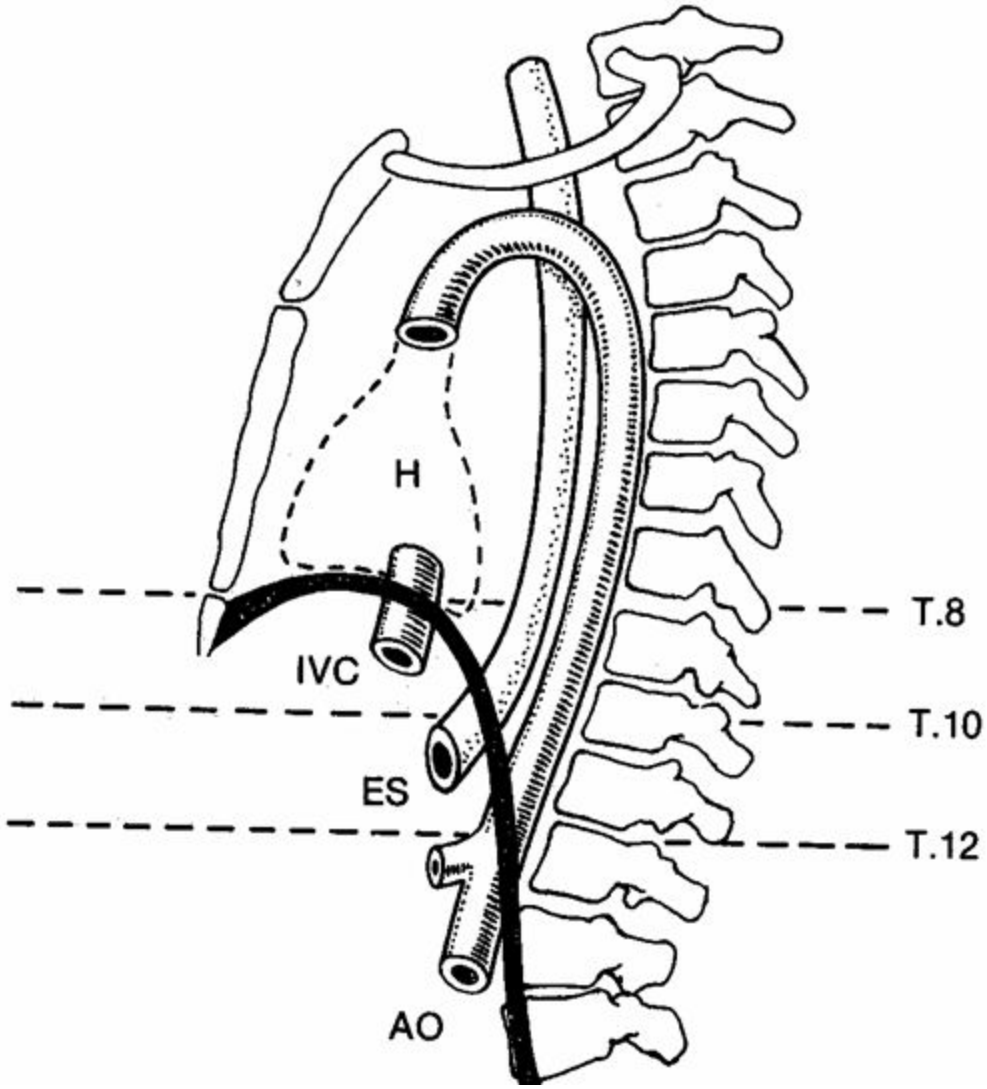
Right Crus L1-L3
Left Crus L1-L2

Vertebral fibrous arches

Median arcuate ligament
  Aorta
Medial arcuate ligament
  Psoas major
Lateral arcuate ligament
  Quadratus lumborum

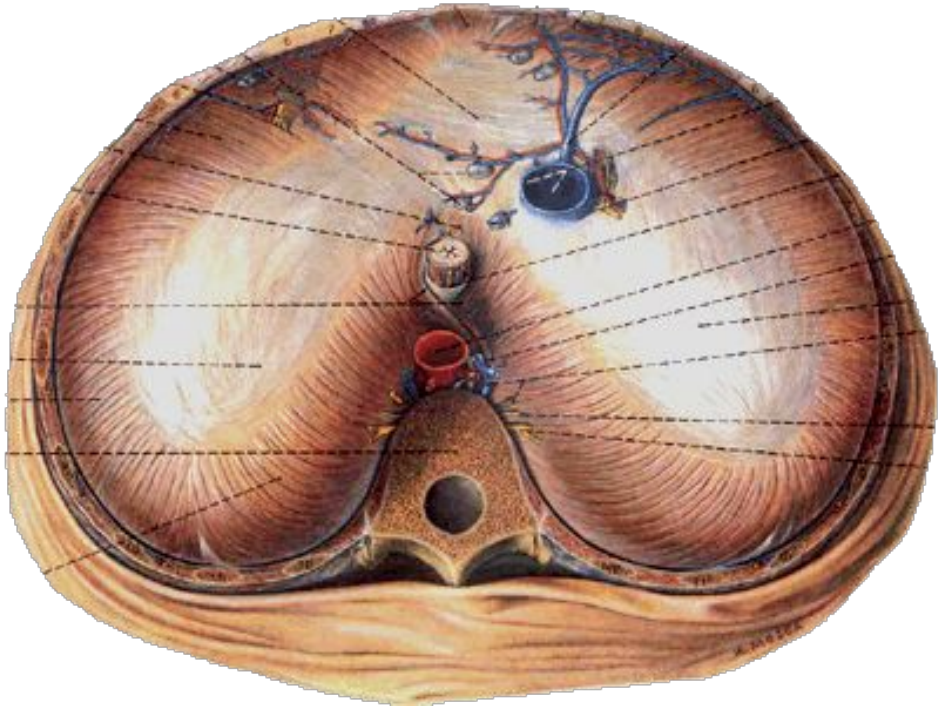


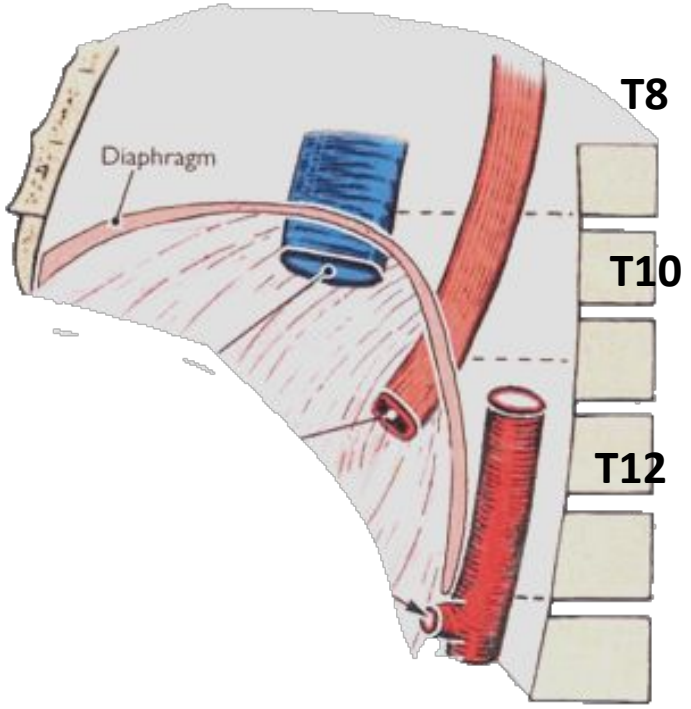
Side view to see curvature of diaphragm.



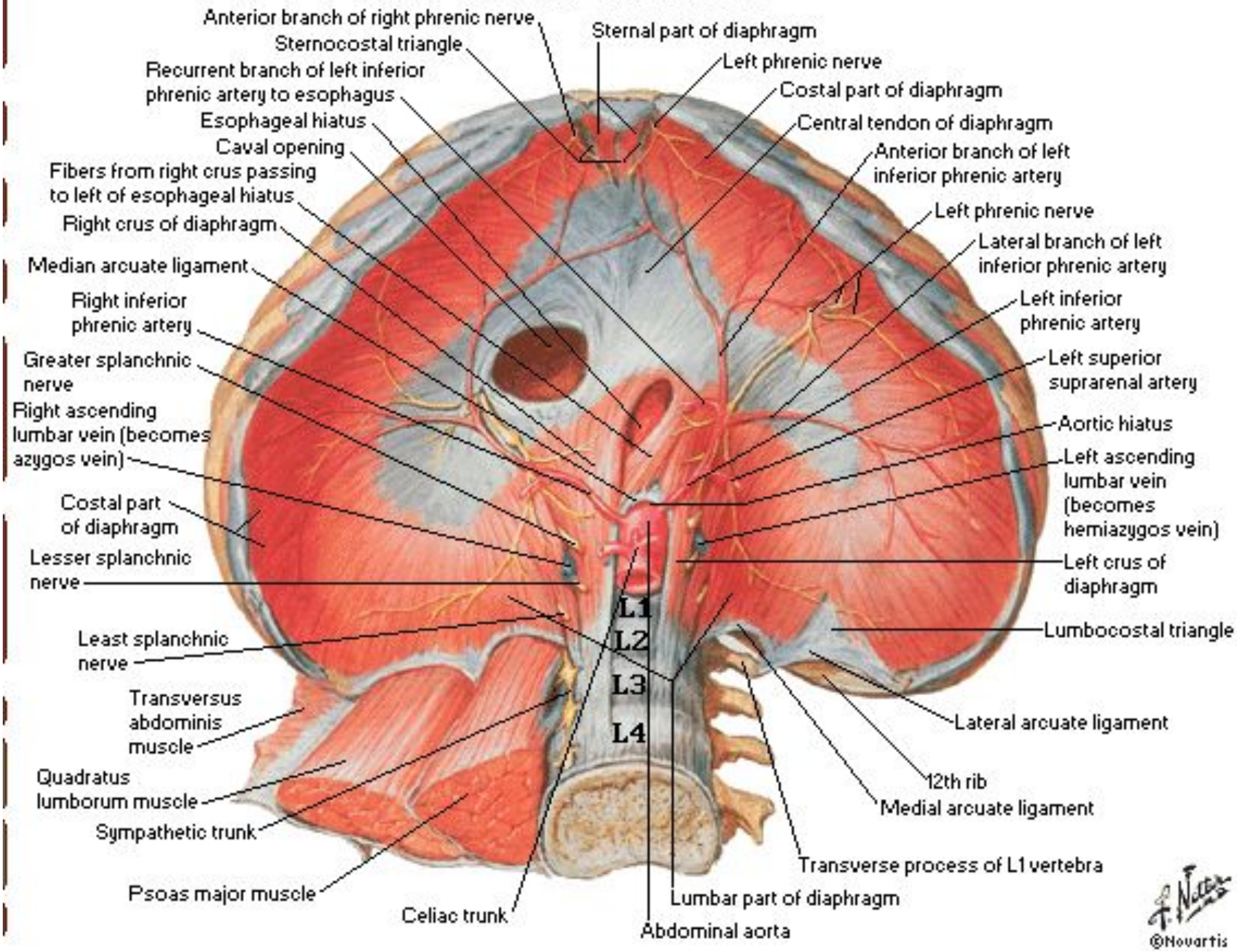
Openings in the diaphragm

- **Aortic hiatus** □ lies anterior to the body of the 12th thoracic vertebra between the crura. It transmits the aorta, thoracic duct
- **Esophageal hiatus** □ for esophagus and vagus nerves at level of T10.
- **Vena cava foramen** □ for inferior vena cava, through central tendon at T8 level



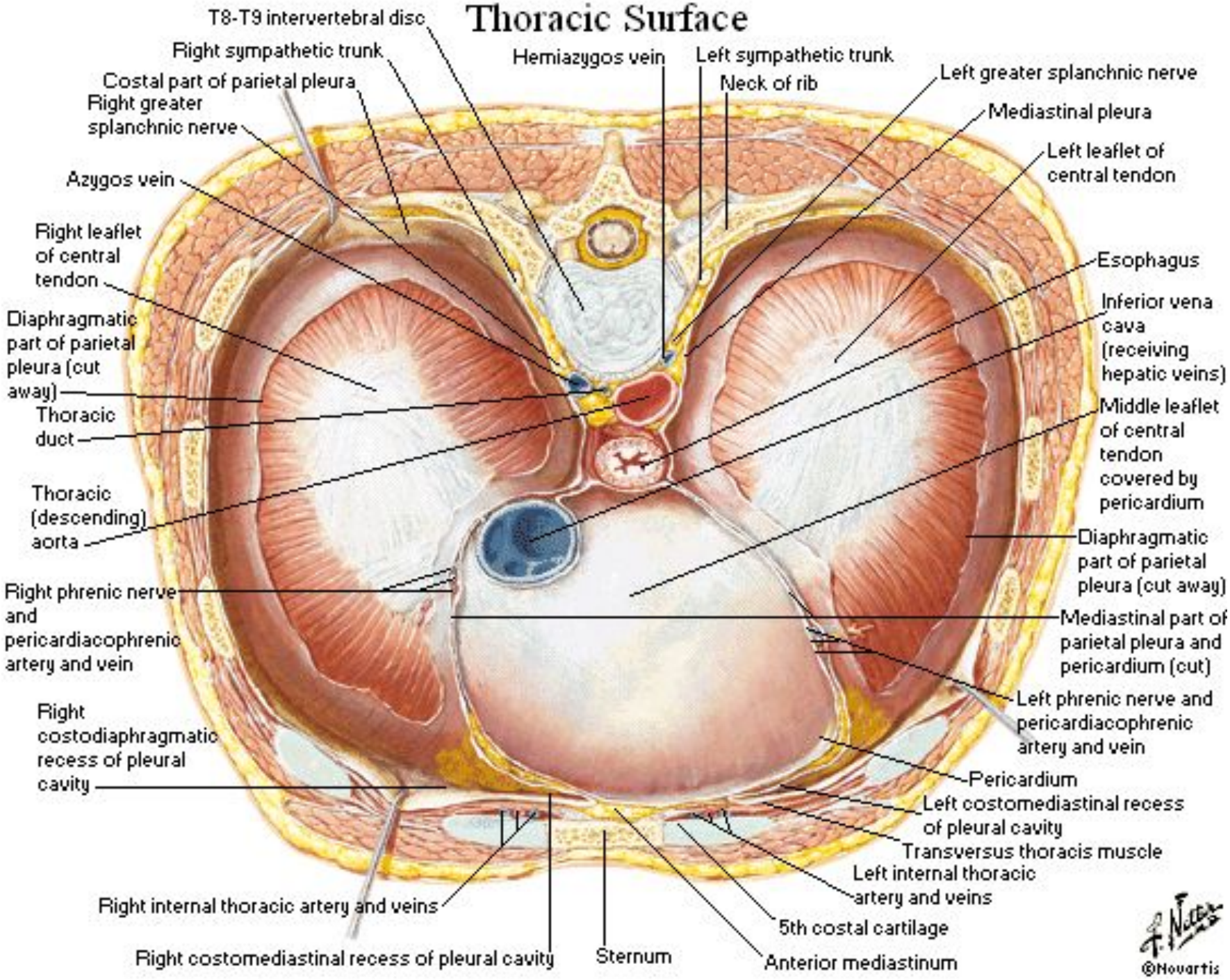


Abdominal Surface



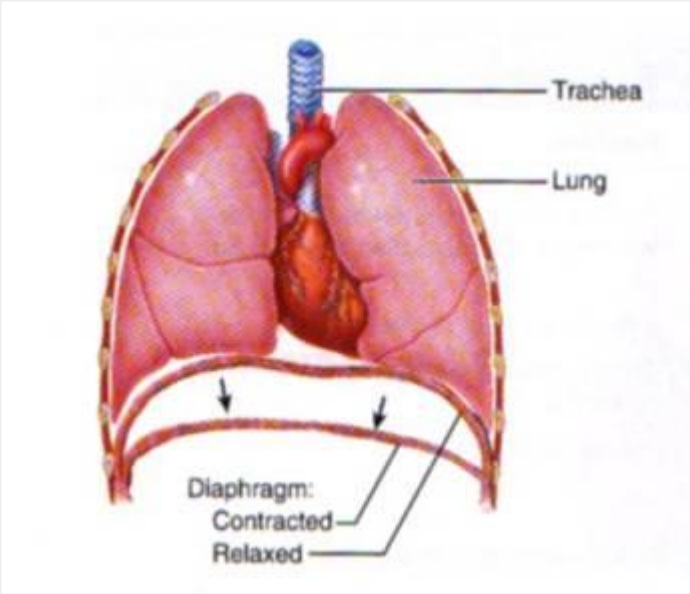
©Novartis

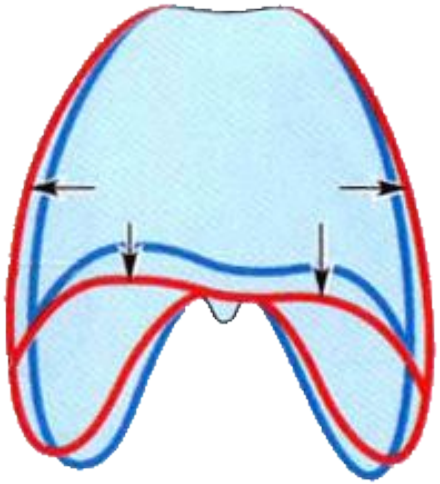
Thoracic Surface



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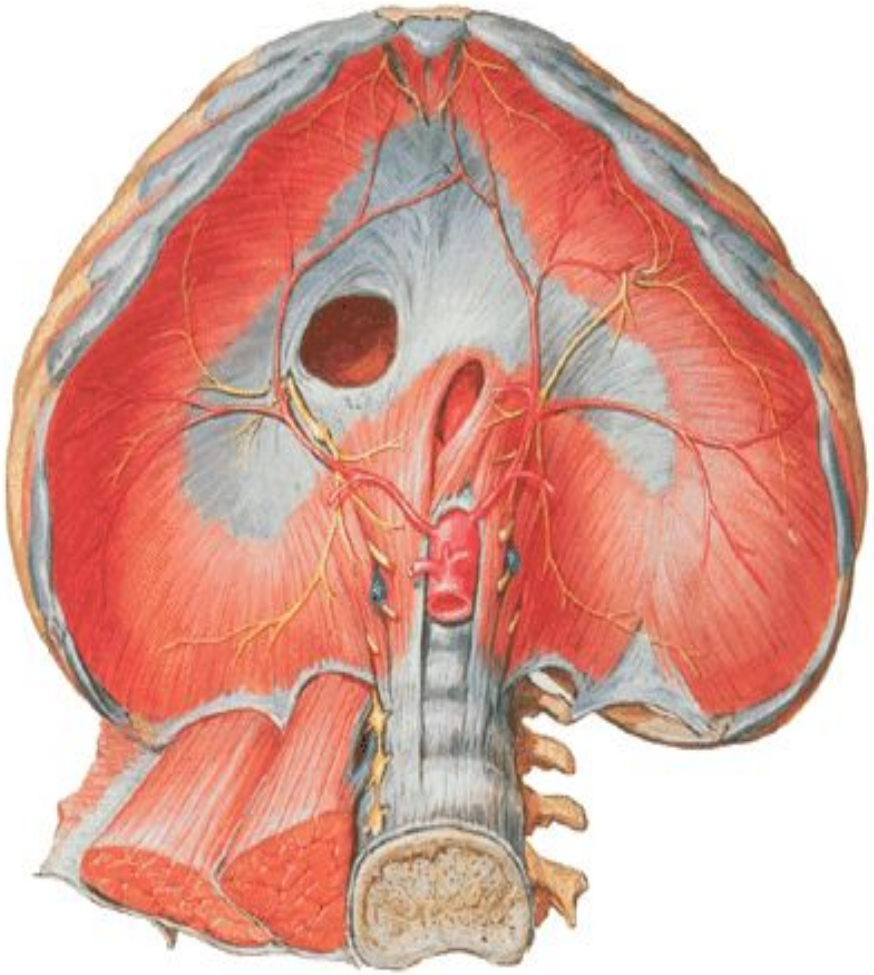
Action of the Diaphragm

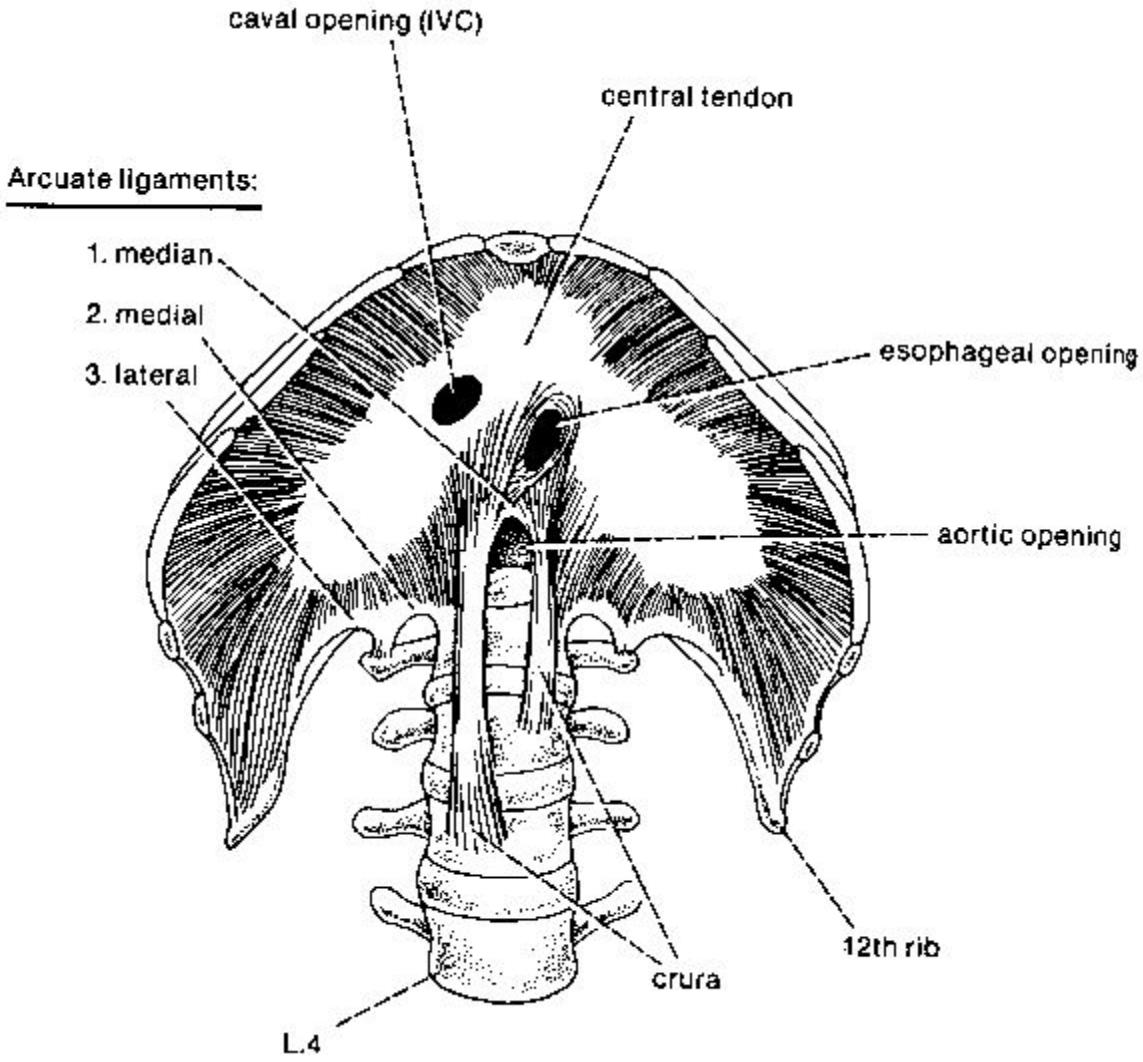




- **Primary muscle of respiration (involuntary)**
  - Contraction during inspiration
    - Increases volume of thoracic cavity
    - Decreases pressure of thoracic cavity
    - Air moves into lungs (high □ low pressure)
- **Forced contraction (voluntary)**
  - Used for defecation, urination, labor
    - Decreases volume of abdominal cavity
    - Increases pressure in abdominal cavity
    - Pushes on abdominal organs to move contents out

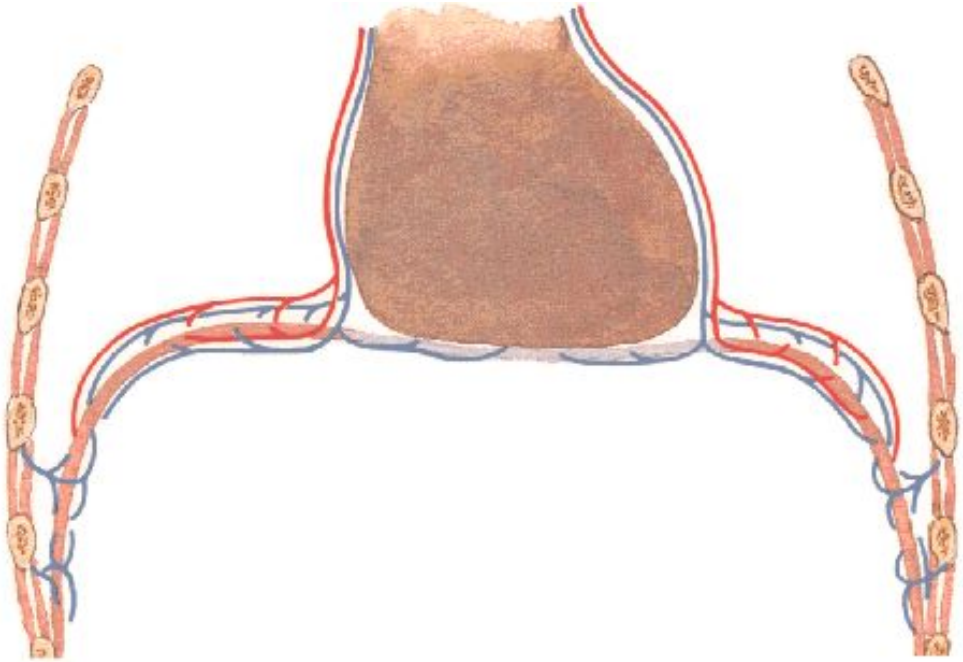
- **Blood supply ~ superior surface**
- Superior phrenic artery (thoracic aorta)
- Musculophrenic and pericardiophrenic arteries (internal thoracic artery)
- **Blood supply ~ inferior surface**
- Inferior phrenic artery (abdominal aorta)

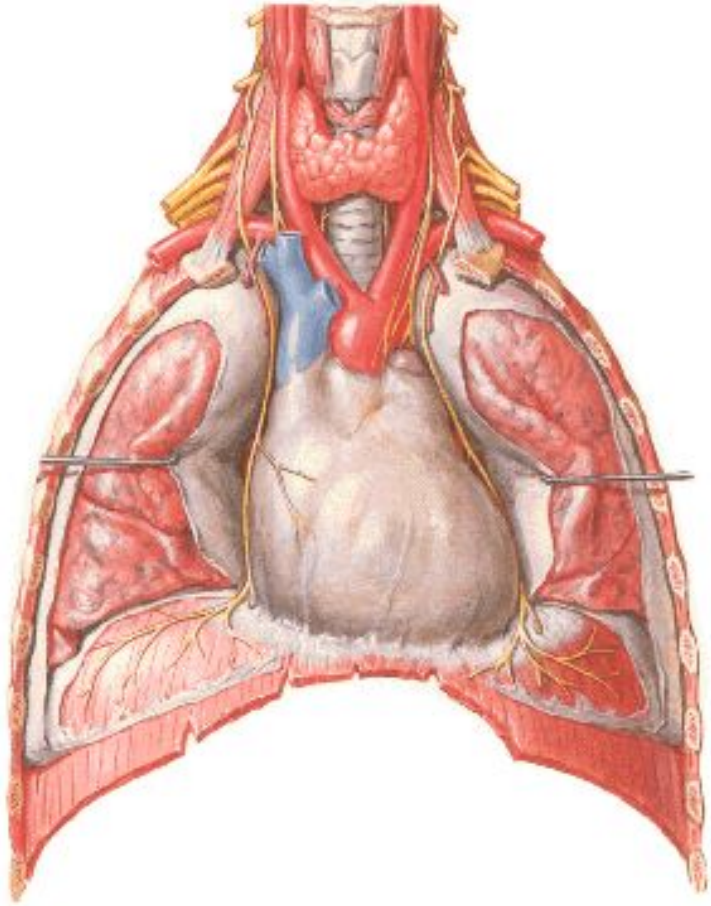




- **Derived from hypaxial musculature of cervical segments.**
- **So motor innervation is from cervical segmental nerves: right and left phrenic nerves (C3,4,5).**

- **Innervation**
  - Motor supply ~ phrenic nerve

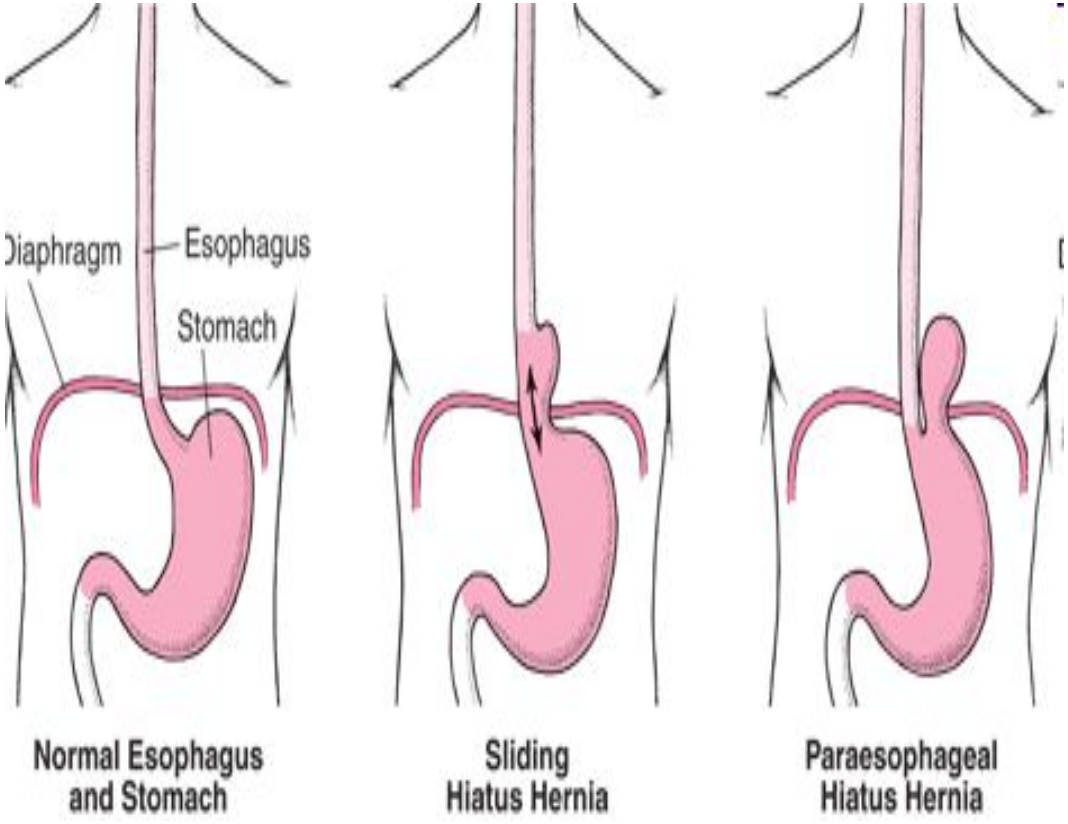




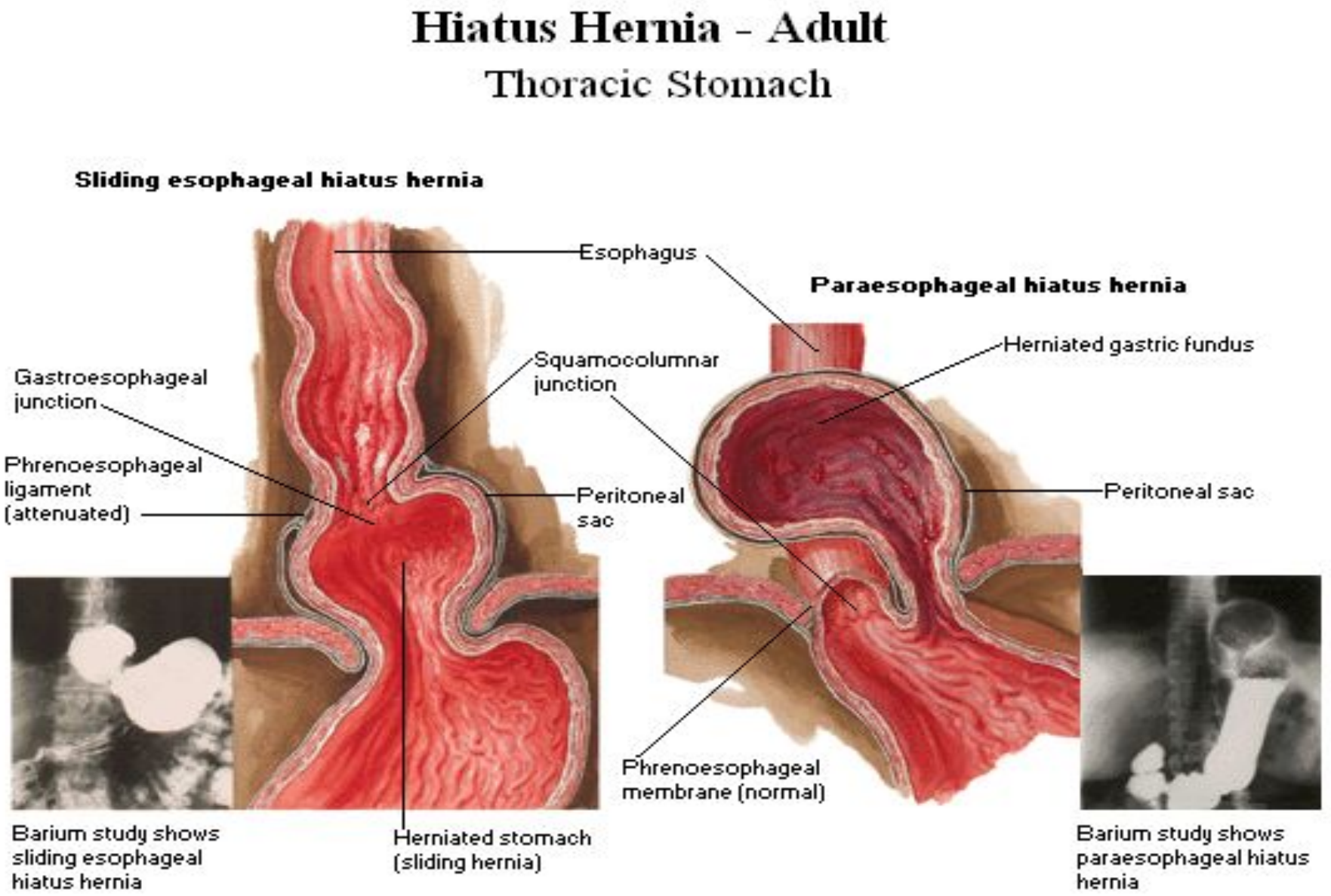
Clinical correlates

Diaphragmatic Hernia:

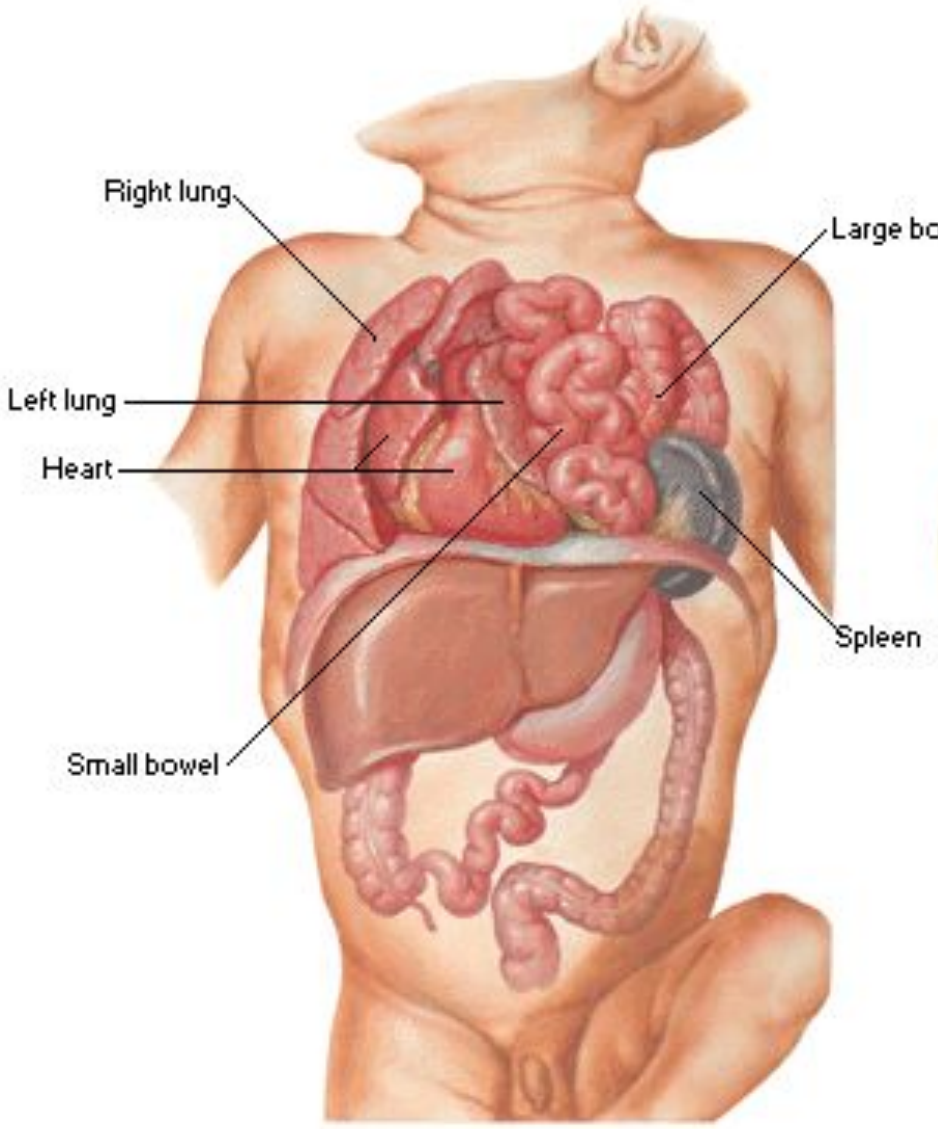
1) **Congenital**
   Failure of pleuroperitoneal membrane development is most common cause.
2) **Acquired**
   Most common is the ***Sliding*** type of hiatus hernia, through the esophageal opening. In this esophagogastric junction rises up in the thorax.
   Very rare variety is ***Rolling*** type, here esophagogastric junction remains in abdomen.



Acquired Esophageal Hernia

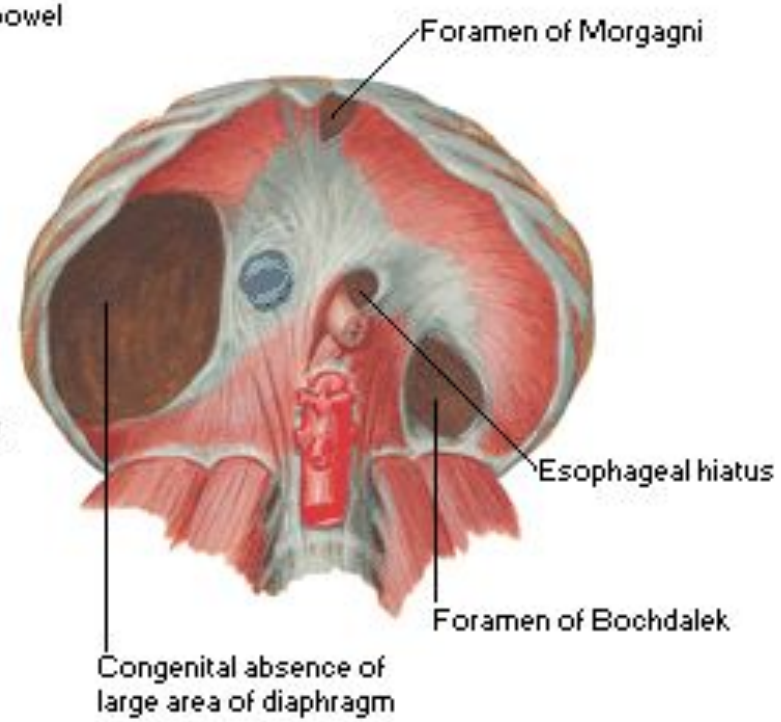


Diaphragmatic Hernia - Congenital



Herniation of abdominal viscera into thorax.
Displacement and compression of thoracic viscera

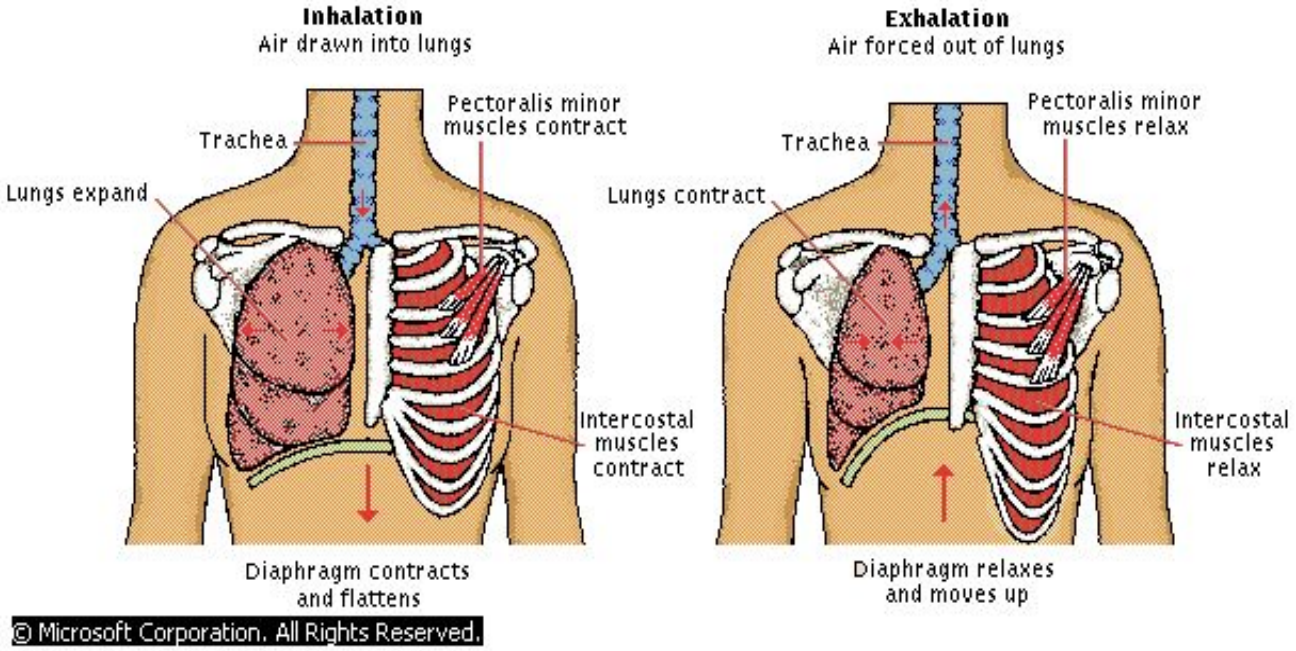
Sites of diaphragmatic hernias



THORACIC MOVEMENT

PRINCIPLE OF THORACIC MOVEMENTS

- The lungs expand during inspiration and retract during expiration
- These movements are governed by the following two factors.
  - Alterations in the capacity of the thorax
  - Elastic recoil of the pulmonary alveoli and of the thoracic wall



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